ICT305 Topics in IT

Topic 7
Mobile Cloud Computing

CIHE Australia

Agenda

- Motivation
- Definition of Mobile Cloud Computing

Motivation

Motivation

- Mobile devices (e.g., smartphone, tablet pcs, etc) are increasingly becoming an essential part of human life,
- Dream of "**Information at your fingertips anywhere anytime",**
- Mobile devices still lack in resources compared to a conventional information processing device such as PCs and laptops

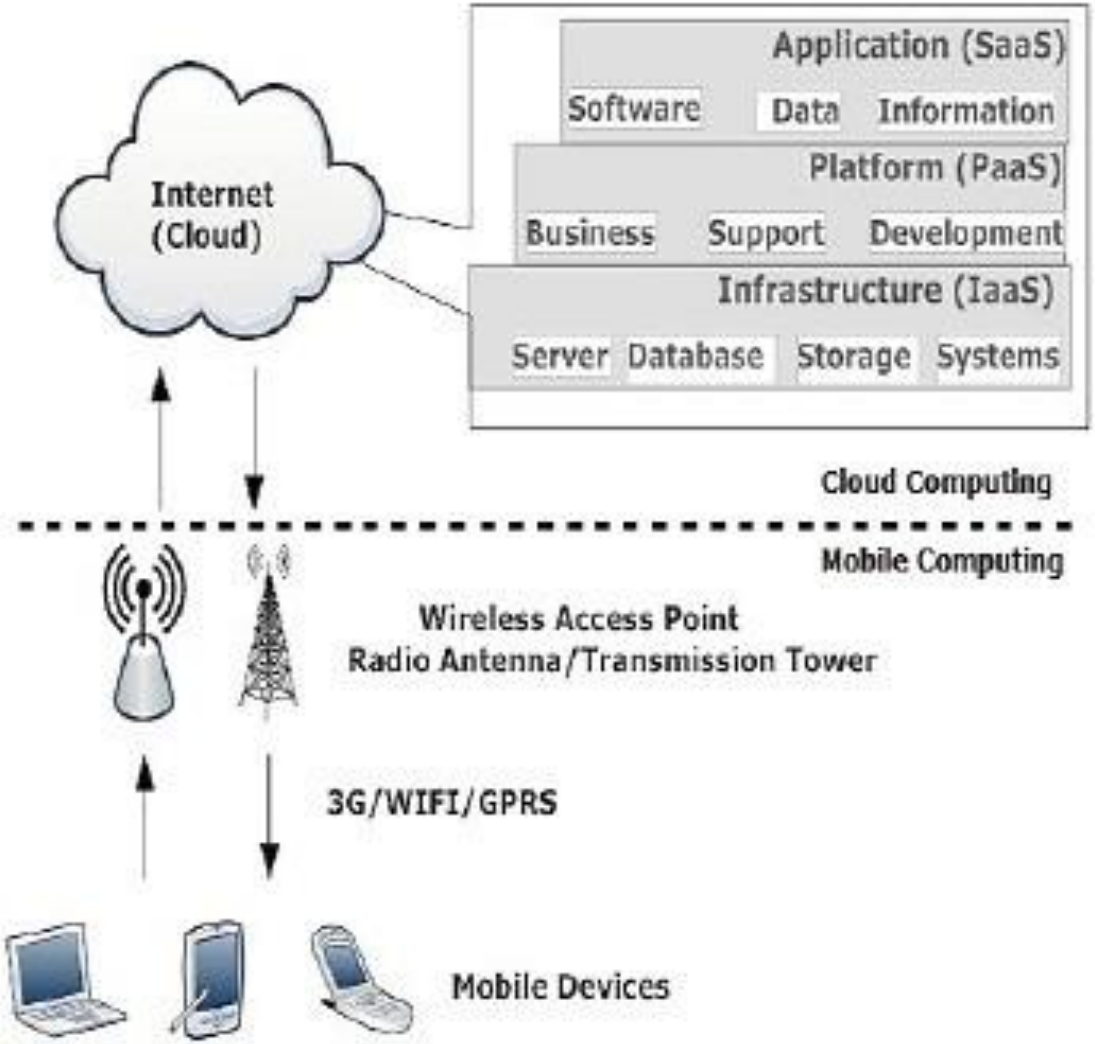
Soultion

› Mobile Cloud Computing (MCC)

Definitions

› Cloud Computing

› Mobile Network

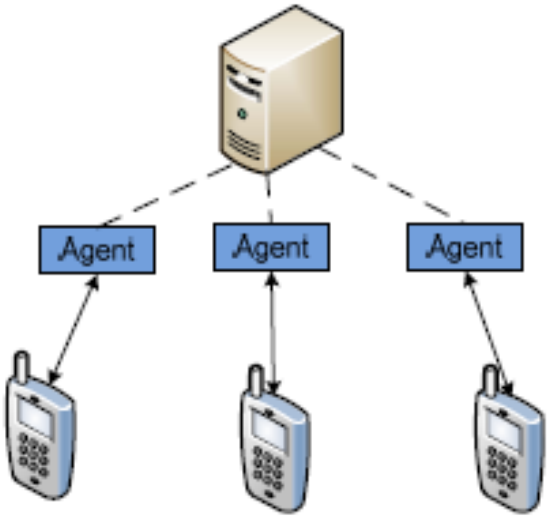


Definition

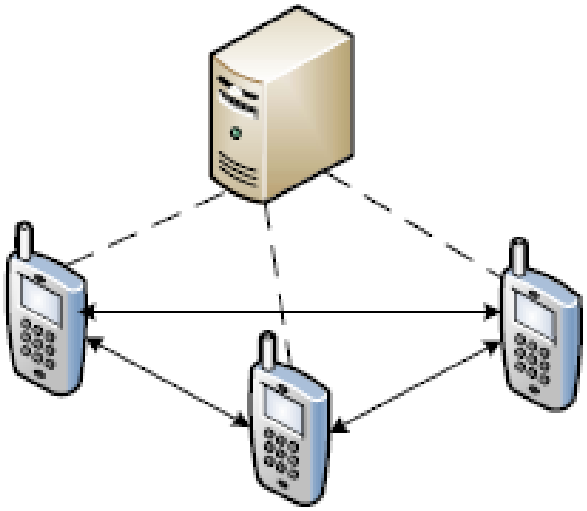
- **Mobile Cloud Computing (MCC)** at its simplest, refers to an infrastructure where both the data storage and the data processing happen outside of the mobile device.
- **Mobile cloud applications** move the computing power and data storage away from mobile phones and into the cloud, bringing applications and mobile computing to not just smartphone users but a much broader range of mobile subscribers.

Architectures of MCC

Agent-client scheme



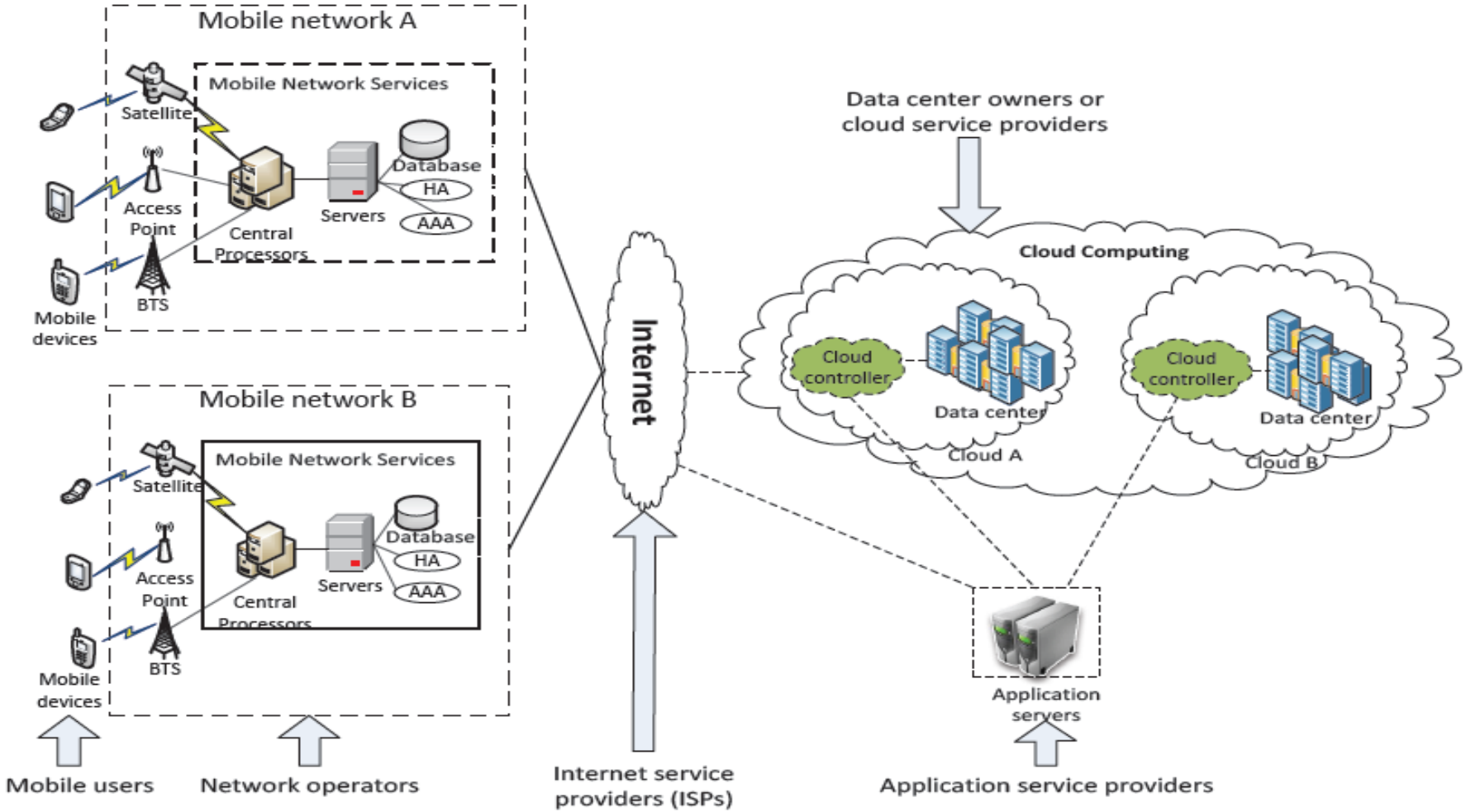
Collaborated scheme



Architectures of MCC

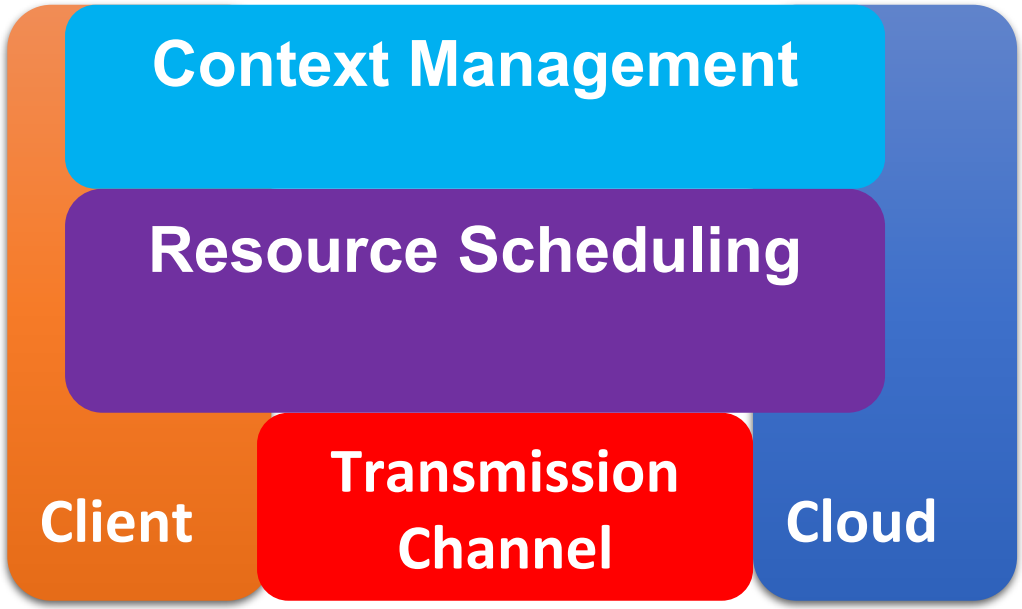
- **Agent-client scheme**
  - In this scheme, cloud side provides overall resource management for mobile devices, to help to overcome limitations of mobile devices in particular of the processing power and data storage.
  - As is shown in Figure of left side, cloud side generate agent for each device.
  - Mobile device communicate with its agent to contact with other entities outside this domain.
- **Collaborated schemes**
  - Regard device as a part of cloud.
  - This approach utilize remain resource of mobile device.
  - The function of cloud server may be the controller and scheduler for collaboration among devices.

Architectures of MCC



Protocol model

- The concept model of cloud computing cannot be directly applied to the MCC.
- **MCC** focuses on the connection between client and cloud, which may differ from common features of cloud computing.



Context management

- **Context Enabled features** of mobile device allow us to ascertain additional information from the computing device itself without the need for explicit user input.
- Two major classes of contexts:
  - *Spatial Context*: contexts that are based on position, proximity
  - *Social context*: contexts that have been explored in social network analysis threads.

Resource scheduling

Resource scheduling components address the schedule of resource, such as computing resource and storage resource.

- **Assign** the appropriate pricing mechanism to maximize the revenue of mobile cloud computing systems and provide incentives for mobile users,
  - *considering different cases*, e.g., resource may be stable; but applications may transmit to other places.

Resource Scheduling: Round Robin Algorithm

Example 1: Round Robin (RR) Scheduling Algorithm

- **Scenario:**
  - You need to schedule tasks on three cloud servers.
- **Round Robin Scheduling:**

1. Initialization:

1. Servers A, B, and C are all initially idle.

2. Task Assignment:

1. Task 1 is assigned to Server A.
2. Task 2 is assigned to Server B.
3. Task 3 is assigned to Server C.
4. Task 4 is assigned to Server A (again, after all servers have been assigned one task).
5. Task 5 is assigned to Server B.
6. Task 6 is assigned to Server C.

Resource Scheduling: Genetic Algorithm

Example 2: Genetic Algorithm (GA) for Task Scheduling

- **Scenario:**
  - You need to schedule four tasks on three cloud servers.
  - Each task has a different size and requires different amounts of resources.
  - **The objective** is to minimize the overall execution time and energy consumption.
- **Task Details:**

Task	Size (in Units)	Server A Time (ms)	Server B Time (ms)	Server C Time (ms)
T1	4	20	25	15
T2	3	10	15	10
T3	5	25	30	20
T4	2	15	20	10

Resource Scheduling: Genetic Algorithm

GA Initialization:

1. Chromosome Representation:

- Each chromosome represents a possible task-to-server assignment.
- For example, a chromosome "A-B-C-A" means Task 1 is assigned to Server A, Task 2 to Server B, Task 3 to Server C, and Task 4 to Server A.

2. Initial Population:

- Generate an initial population of chromosomes randomly. Example:
    1. Chromosome 1: A-B-C-A
    2. Chromosome 2: B-A-C-B
    3. Chromosome 3: C-B-A-C

3. Fitness Function:

- The fitness function evaluates the total execution time and energy consumption for each chromosome.
- The objective is to minimize these values.

Resource Scheduling: GA

Fitness Calculation Example:

- Let's calculate the fitness for Chromosome 1 (A-B-C-A):
  - **Task 1 on Server A**: Time = 20 ms
  - **Task 2 on Server B**: Time = 15 ms
  - **Task 3 on Server C**: Time = 20 ms
  - **Task 4 on Server A**: Time = 15 ms
- **Total Time (Makespan):**
  - Makespan = max(20 + 15, 15, 20, 20) = 35 ms
- **Fitness (Inverse of Makespan):**
  - Fitness = 1 / 35 = 0.0286

Resource Scheduling: GA

Selection, Crossover, and Mutation:

4. **Selection**: Select the best-performing chromosomes (e.g., those with the highest fitness scores) for reproduction.
5. **Crossover**: Combine parts of two parent chromosomes to create new offspring. For example:
   a. Parent 1: A-B-C-A
   b. Parent 2: C-B-A-C
   c. Offspring: A-B-A-C
6. **Mutation**: Occasionally, mutate a gene in a chromosome to maintain genetic diversity. For example, change "A-B-A-C" to "A-C-A-C."
7. **Iteration**: Repeat the process of selection, crossover, and mutation for several generations until the algorithm converges to an optimal or near-optimal solution.

- **Final Solution:**
  - After several iterations, the GA might converge to a chromosome like "A-C-A-B," which provides the best balance of execution time and energy consumption.

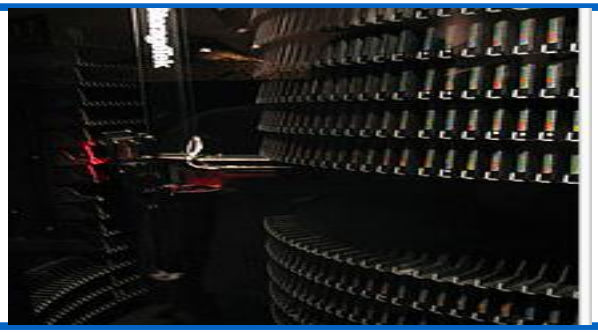
Context Management and major approaches

- **Context Enabled features** of mobile device allow us to ascertain additional information from the computing device itself without the need for explicit user input.
  - Two major classes of contexts: Social Context and Spatial context.
- **Application partition** and **offloading technology** play an important role for the implementation of elastic applications.
  - **Application partition** decompose complex workload to atomic ones, thus can be processed concurrently.
  - **Offloading application** can free burden of mobile devices.

Advantages of MCC

Extending battery lifetime

Improving data storage capacity and processing power



Improving reliability

How MCC Can Extend Battery Lifetime?

Challenges:

- Battery is one of the main concerns for mobile devices,
- Traditional approaches need to changes the structure of mobile devices.
- The additional cost for the end mobile users is not appealing in wireless networks.

MCC's solution:

- **Computation offloading technique:**
  - Immigrate the large computations and complex processing from resource-limited devices (i.e., mobile devices) to resourceful machines (i.e., servers in clouds).
- This avoids taking a long application execution time on mobile devices which results in large amount of power consumption.

How MCC Can Improve Storage Capacity?

Challenges

- Users need more and more capacity for saving the essential information on mobile devices,
- Need to change the device,
- More capacity, more weight

MCC's solution

- MCC is developed to enable mobile users to store/access the large data on the cloud through wireless networks,
- Examples of existing services:
  - Amazon Simple Storage Service (Amazon S3),
  - Image Exchange,
  - Flickr, ShoZu.

How MCC Can Improve Reliability?

Challenges

- Users need reliable backup for their information,
- Lack of data security model for both service providers and users in existing mobile users,

MCC's solution

- Storing data or running applications on clouds is an effective way to improve the reliability since the data and application are stored and backed up on a number of computers.

Other advantages of MCC

- Dynamic provisioning:
  - A fine-grained, self-service basis is a flexible way to run applications without advanced reservation of resources.
- Scalability:
  - Service providers can easily add and expand an application and service.
- Multi-tenancy:
  - Service providers can share the resources and costs to support a variety of applications and large number of users.
- Ease of integration:
  - Multiple services from different service providers can be integrated easily

Applications of MCC

- Mobile commerce,
- Mobile healthcare,
- Mobile learning,
- Mobile Gaming.

Mobile commerce

› **Mobile commerce** (m-commerce) is a business model for commerce using mobile devices; for example, commbank mobile app.

Challenges of m-commerce applications (e.g., low network bandwidth, high complexity of mobile device configurations, and security).

M-commerce applications are integrated into cloud computing environment

Mobile learning

- Traditional m-learning applications have **limitations** in terms of
  1. **High cost** of devices and network,
  2. **Low** network transmission rate,
  3. **Limited** educational resources
- **Cloud-based m-learning** applications are introduced to solve these limitations.
- **Utilizing a cloud** with the large storage capacity and powerful processing ability,
  - the applications provide learners with much richer services in terms of data (information) size, faster processing speed, and longer battery life.

Mobile-healthcare

- Comprehensive health monitoring services,
- Intelligent emergency management system

- Health-aware mobile devices detect pulse-rate,
- Pervasive access to healthcare information,

- Pervasive lifestyle incentive.

Mobile Gaming

- **Mobile game** (m-game) is a potential market generating revenues for service providers.
- M-game can completely offload game engine requiring large computing resource (e.g., graphic rendering) to the server in the cloud, and gamers only interact with the screen interface on their devices.

Issues in Mobile Communication Side

Network latency and limited bandwidth

Availability

Heterogeneity

Low Bandwidth Solutions

Share the limited bandwidth among mobile users who are located in the same area (e.g., a workplace, a station, and a stadium) and involved in the same content (e.g., a video file).

X. Jin, etal, "Cloud Assisted P2P Media Streaming for Bandwidth Constrained Mobile Subscribers," (ICPADS), pp. 800, January 2011.

Data distribution policy which determines when and how much portions of available bandwidth are shared among users from which networks (e.g., WiFi and WiMAX).

E. Jung, etal "User-profile-driven collaborative bandwidth sharing on mobile phones" in MCS, no. 2, 2010.

Availability Solutions

- **Finding stable neighbour WiFi multi-hop networking system**

- G. Huerta “A virtual cloud computing provider for mobile devices,” in MCS, 2010.
- L. Zhang,, “WiFace: a secure geosocial networking system using WiFi-based multi-hop MANET,” in MSC, 2010.

Issues in Computing Side

- Computing offload
- Security
- Enhancing the efficiency of data access
- Context aware mobile cloud services

Issue and Solutions in Computing Offload

- Offloading in the statistic environment is not always the efficient way to save energy
  - for small calculation and depending on the transmission technology;
  - Tradeoff between communication and computation cost.

Energy-Efficient Task Offloading in MCC

Scenario: A mobile device is connected to a cloud server and has to execute a computational task. The task can either be executed locally on the mobile device or offloaded to the cloud server. The goal is to determine the most energy-efficient strategy while considering the time delay.

1. **Task Size**:
    1. Task size = 2 MB (Megabytes)
2. **Mobile Device Specifications**:
    1. CPU processing power = 500 MIPS (Million Instructions Per Second)
    2. Energy consumption rate when processing locally = 0.5 Joules/sec
    3. Network bandwidth = 2 Mbps (Megabits per second)
    4. Energy consumption rate during data transmission = 0.3 Joules/sec
3. **Cloud Server Specifications**:
    1. Cloud server CPU processing power = 1000 MIPS
    2. Round-trip network latency = 50 ms (milliseconds)
4. **Task Instructions**:
    1. Number of instructions = 200 million

Calculations:

- We'll calculate the following for both local execution and cloud offloading:
    1. **Execution Time**: Time taken to execute the task.
    2. **Energy Consumption**: Energy consumed during execution.

Offloading

1. Local Execution on Mobile Device

Execution Time (Local):

- **Time to Execute Task**:

$$\text{Time (Local)} = \frac{\text{Number of Instructions}}{\text{Mobile Device CPU Power}} = \frac{200 \text{ million instructions}}{500 \text{ MIPS}} = \frac{200}{500} = \text{ 0.4 (s)}$$

Energy Consumption (Local):

- **Energy Consumed**:

$$\text{Energy (Local)} = \text{Energy Rate (Local)} \times \text{Time (Local)} = 0.5 \text{ Joules/sec} \times 0.4 \text{ sec}$$

Offloading

2. Offloading Execution to Cloud Server

Data Transmission Time:

- **Time to Transmit Data to Cloud**:

$$\text{Time (Transmission)} = \frac{\text{Task Size} \times 8}{\text{Network Bandwidth}} = \frac{2\text{ MB} \times 8\text{ bits}}{2\text{ Mbps}} = \frac{16\text{ Mbits}}{2\text{ Mbps}} = 8 \text{ seconds}$$

Execution Time (Cloud):

- **Time to Execute Task on Cloud**:

$$\text{Time (Cloud Execution)} = \frac{\text{Number of Instructions}}{\text{Cloud CPU Power}} = \frac{200\text{ million instructions}}{1000\text{ MIPS}} = 0.2 \text{ seconds}$$

Total Time (Offloading):

- **Total Time for Offloading**:

$$\text{Total Time (Offloading)} = \text{Time (Transmission)} + \text{Time (Cloud Execution)} + \text{RTT}$$

$$= 8\text{ seconds} + 0.2\text{ seconds} + 0.05\text{ seconds} = 8.25\text{ seconds}$$

Offloading

Energy Consumption (Offloading):

- **Energy for Data Transmission**:

$$\text{Energy (Transmission)} = \text{Energy Rate (Transmission)} \times \text{Time (Transmission)} =$$

- **Total Energy Consumption (Offloading)**:

$$\text{Energy (Offloading)} = \text{Energy (Transmission)} + \text{Energy (Cloud Execution)}$$

Since energy consumption on the cloud side is usually negligible for the mobile device:

$$\text{Energy (Offloading)} = 2.4 \text{ Joules}$$

Issues in Computing Side

- Security
  - Security for mobile users,
    - Security for mobile applications,
    - Privacy
  - Security of data on cloud
    - Integrity
    - Authentication,

Summary: Mobile cloud computing service convergence

- The development and competition of cloud service providers can lead to the fact that in the near future these services will be differentiated according to the types, cost, availability and quality.
- A single cloud is not enough to meet mobile user's demands.
- The new scheme is needed in which the mobile users can utilize multiple cloud in a unified fashion
- The **mobile sky computing**, will enable providers to support a cross-cloud communication and enable users to implement mobile services and applications.

Top 10 Mobile cloud computing service providers

- https://blog.back4app.com/mobile-cloud/